Sheel Patel

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github.com/Sheel34 | Portfolio | Open to Relocation

Achievements

National Winner — Limitless Youth India Hackathon **March 2025**

- Ranked **1st out of 10,000+ teams** nationwide in Science & Technology; directly invited by the organising committee with no open application
- System evaluated by a jury comprising the **Chairman of ISRO**, NITI Aayog members, and DRDO technologists; received recognition from the **Prime Minister of India**
- Project cited as a breakthrough in real-time robotic control systems for defence and hazardous-environment applications

Projects

BHUVAN — Terrain Intelligence System | *Python, React, Three.js, FastAPI, NumPy, OpenCV* | Live Demo **Present**

- Engineered a multi-layer **terrain-risk pipeline** computing slope (gradient), surface roughness (sliding-window std. dev.), curvature (Laplacian), and a hillshading proxy (surface-normal dot product), combined into a weighted hazard score per grid cell
- Built a **FastAPI REST backend** exposing /analyze, /analyze-upload, and /samples endpoints; ingests PNG, JPEG, and GeoTIFF (Rasterio) files up to 50 MB, normalises spatial grids, and returns fully typed JSON payloads covering hazard, traversability, and ranked landing zones
- Developed a React + **Three.js (WebGL)** 3D mission-control dashboard with configurable terrain overlay layers, interactive landing zone selection, real-time lander physics simulation, and downloadable mission reports
- Validated against procedural analogues of Mars Jezero Crater, Gale Crater, and Lunar South Pole elevation datasets; maintained module-level technical documentation throughout

VENOM — EOD Robotic Arm | *C++, Python, ESP32, FreeRTOS, OpenCV* | GitHub | Demo **Feb 2025**

- Engineered a **5-DOF teleoperated robotic arm** for Explosive Ordnance Disposal that replicates human hand kinematics in real time, reducing simulated task completion time by 62%
- Designed a custom operator glove integrating 10+ flex sensors and an MPU-6050 IMU on ESP32; achieved **~20 ms end-to-end wireless latency** over the ESP-NOW protocol
- Implemented **FreeRTOS multithreaded firmware** for deterministic task scheduling, enabling parallel control of 12+ servos; inverse kinematics solver achieving sub-centimetre positioning accuracy
- Attained 74% improvement in EOD simulation success rates; drew active interest from **3+ national defence research institutions**

Technical Skills

Languages: Python, JavaScript, C, C++, Java
Data & Geospatial: NumPy, Pandas, OpenCV, Rasterio, GeoTIFF, spatial/terrain analysis, statistical signal processing, SQL
Web & Visualisation: React (Vite/PWA), Three.js, WebGL, FastAPI, REST APIs, JSON telemetry pipelines, 3D rendering
Embedded & Robotics: ESP32, FreeRTOS, PWM, I2C, SPI, ESP-NOW, Inverse Kinematics, Servo Control, Sensor Fusion, MPU-6050
Tools: Git, Linux, Docker (familiar), Azure (familiar), ServiceNow, Agile

Certifications

ServiceNow Certified System Administrator (CSA) | ServiceNow **March 2026**

- Certified in platform administration, automated workflow design, and ITSM configuration at enterprise scale

ServiceNow Certified Application Developer (CAD) | ServiceNow **June 2026**

- Certified in application development, scripting, UI policies, business rules, and custom app deployment on the Now Platform

Education

Parul University **Vadodara, Gujarat**
B.Tech — Artificial Intelligence & Data Science | *CGPA: 7.36* *July 2023 – May 2027*

SSV 2 School **Vadodara, Gujarat**
GHSEB — Physics, Chemistry, Mathematics (PCM) — 61.5% *June 2021 – April 2023*